

FUD CLINIC SYSTEM UPGRADE

Printed Project Requirements, Feature Classification & Implementation Plan

Scope: The clinic system is designed for the wider FUD clinic/hospital environment, not only for students. It should support students, staff, patients, pregnant women, doctors, nurses, pharmacy staff, laboratory staff, receptionists and other authorised users.

1. Clean Printed Version of the Forwarded Notes

1. Card number
2. Appointment
3. Test result on portal
4. Doctor
5. Pharmacy
6. Report
7. Emergency
8. Registration number, faculty and department
9. Linking the project with email alerts/notifications
10. Finding an expert in the medical field
11. For people who do not have mobile phones: provide access through clinic computers or the receptionist
12. Number of people who booked to see a doctor and their information
13. Pregnancy section
14. Questionnaire
15. System verification (model)
16. Program testing
17. Creating an account
18. Informing patients who did a test when their test result is ready

2. Main Algorithm Approach: Divide and Conquer

The best way to organise this project is to use a Divide-and-Conquer approach: break the large clinic system into smaller modules, complete and test each module, and then integrate the modules into one system. This makes the project easier to develop, test, explain and maintain.

Module	What it handles
A. User & Registration	Account creation, card number, registration number, faculty, department and patient profile.
B. Appointment	Doctor selection, available dates/times, booking, cancellation/rescheduling and appointment status.
C. Doctor	Doctor profile, specialty, availability, patient list and consultation information.
D. Laboratory/Test	Test request, sample/status tracking, result entry and result availability.
E. Pharmacy	Prescription, medicine records and pharmacy workflow.
F. Reports	Patient reports, consultation records and authorised report viewing.
G. Emergency	Emergency request, priority handling and quick access to emergency support.
H. Pregnancy/Maternity	Pregnancy-related registration, appointments, follow-up information and maternity records.
I. Questionnaire	Patient questions/assessment forms and submission of responses.
J. Notifications	Email/SMS/in-system alerts for appointments and ready test results.

Module	What it handles
K. Expert Finder	Search for an appropriate medical specialist by area of expertise.
L. Reception/No-Phone Access	Receptionist or clinic-computer access for patients without smartphones.
M. Administration & Verification	User roles, permissions, system verification, security checks and audit records.

3. Feature Classification

A. START WITH — SIMPLEST / FOUNDATION FEATURES

1. Create an account
2. Registration details: card number, registration number, faculty and department
3. Basic patient profile
4. Questionnaire
5. Basic doctor profiles
6. Basic pregnancy section structure

B. NEXT — MEDIUM COMPLEXITY FEATURES

7. Appointment booking
8. Doctor availability and appointment status
9. Pharmacy/prescription records
10. Reports and authorised patient records
11. Test request and test-result display
12. Number of booked patients and appointment information

C. TIME-CONSUMING / HARDER FEATURES

13. Email/in-system alerts and notifications
14. Automatically informing patients when test results are ready
15. Emergency workflow and priority handling
16. Medical expert/specialist finder
17. Receptionist/computer access for patients without mobile phones
18. Pregnancy/maternity workflow with follow-up records
19. Role-based access and system verification/security

D. FINAL PHASE — TESTING & INTEGRATION

20. Program testing: test each module separately.
21. System verification: confirm that modules work together correctly.
22. Integration: connect the completed modules into one clinic system.
23. User acceptance testing: let representative clinic users try the main workflows and report problems.

Recommended development order

Start with account/registration → patient profile → doctor records → appointment → laboratory/test results → pharmacy → reports → pregnancy module → notifications → emergency → expert finder → receptionist/no-phone access → security/verification → full testing.

4. Appointment — How the Workflow Should Work

Step	Process
1	Patient logs in or the receptionist opens the patient's record.
2	Patient selects the service: general consultation, pregnancy/maternity, follow-up, etc.
3	System shows doctors/specialists relevant to that service.
4	System checks the doctor's available dates and time slots.

Step	Process
5	Patient selects an available date and time.
6	System confirms the booking and creates an appointment number/reference.
7	Patient receives an in-system/email alert; the receptionist can also see the booking.
8	Doctor sees the appointment on the doctor's dashboard.
9	After consultation, the appointment is marked completed and the relevant report/record is saved.
10	If necessary, the doctor creates a test request, prescription, follow-up appointment or referral.

Important appointment rules

- A time slot should not be bookable by two patients at the same time.
- The system should show appointment status: Pending, Confirmed, Completed, Cancelled or Rescheduled.
- Patients should be able to see their upcoming appointments.
- Doctors/receptionists should be able to see the day's appointments and patient information according to their permissions.
- For urgent cases, the emergency workflow should not depend on a normal appointment slot.

5. Test Result Workflow

- Doctor/authorised staff requests a test.
- Laboratory records the test and its status.
- Result is entered/verified by authorised laboratory staff.
- System changes status to 'Ready' when appropriate.
- Patient receives a notification that the result is ready.
- Patient logs in to view the result, subject to the clinic's access rules.
- The result remains attached to the patient's medical record for authorised staff.

6. Recommended System Model / Architecture

For a project like this, use an MVC (Model–View–Controller) architecture. It separates the data and business rules from the user interface and the request-handling logic. This fits the modular Divide-and-Conquer plan.

Part	Role in the clinic project
Model	Database/data and business rules: patients, doctors, appointments, tests, results, prescriptions, reports, pregnancy records and notifications.
View	What users see: patient dashboard, doctor dashboard, receptionist dashboard, pharmacy/lab screens and admin pages.
Controller	Receives user actions and connects the View to the Model: booking appointments, submitting questionnaires, requesting tests, etc.

Suggested core database entities

User / Account • Patient • Doctor • Department • Appointment • Medical Record • Pregnancy Record • Questionnaire • Test • Test Result • Prescription • Medicine • Report • Emergency Request • Notification • Specialist/Expertise • Audit Log

7. Important Things Missing / Recommended Additions

- Role-based access control: patient, doctor, nurse, receptionist, laboratory staff, pharmacy staff and administrator should not all have the same access.
- Password reset and account recovery.
- Appointment cancellation/rescheduling.
- Doctor availability/scheduling.
- Patient medical-history record.
- Search and filtering for authorised staff.
- Audit log: record important actions such as who changed a result or appointment.
- Data backup and recovery.
- Privacy/security measures because the system handles medical information.
- Mobile-responsive web interface so it can work on phones and clinic computers.
- Clear emergency contact/action screen.
- Validation and error messages so incorrect information is not saved.

8. Final Project Breakdown

Problem: Clinic users may face difficulty with registration, appointments, test-result access, doctor discovery, pharmacy records, emergency handling, pregnancy/maternity services and communication.

Solution: Build one modular clinic-management system that connects patients, doctors, laboratory, pharmacy, reception and administration while giving patients appropriate access to their information.

Algorithm: Divide and Conquer — divide the large clinic system into smaller modules, build and test them, then integrate them into one complete system.

Architecture/Model: MVC, supported by a well-structured relational database and role-based access control.

First module to build: Account + registration/patient profile. **Most time-consuming:** appointment scheduling, notifications, emergency handling, specialist finder, multi-role access and medical workflows. **Final stage:** integration, system verification and full program/user testing.